

Momentum Homework Set

Instructions

- Show all work clearly.
- Choose a coordinate system and define directions.
- Use vector components where necessary.
- State final answers with correct units and directions.

1 Short Problems

Problem 1 (Impulse)

A 3 kg object is moving to the right at 4 m/s. A constant net force of 6 N acts on it in the same direction for 5 seconds.

Find: The final velocity of the object.

Problem 2 (1D Collision)

A 2 kg cart moving to the right at 5 m/s collides and sticks to a 3 kg cart moving to the left at 1 m/s. Prove that the collision is inelastic.

Find: The final velocity of the combined system.

2 Long Problems

Problem 3 (Explosion in 2D)

A 300 kg object at rest explodes into two pieces.

- Piece 1 has mass 100 kg and moves at 80 m/s at 40° north of east.
- Piece 2 has mass 200 kg.

Find:

1. The velocity components (v_x , v_y) of piece 2.
2. The magnitude of the velocity of piece 2.
3. The direction of motion of piece 2 (angle and quadrant).

Problem 4 (Multi-Physics: Kinematics + Energy + Momentum + Friction)

A 1 kg block starts from rest at the top of a rough incline of height 8 m and angle 30° . The coefficient of kinetic friction between the block and incline is $\mu_k = 0.1$.

At the bottom of the incline, the block moves onto a horizontal frictionless surface and collides elastically with a 2 kg block initially at rest.

After the collision, the 2 kg block slides across a rough horizontal surface with $\mu_k = 0.2$ until it comes to rest.

Find:

1. The acceleration of the 1 kg block as it slides down the incline.
2. The speed of the 1 kg block just before reaching the bottom (use kinematics or energy).
3. The velocities of both blocks immediately after the elastic collision.
4. The distance traveled by the 2 kg block before stopping.

Notes

- Use kinematics or energy for motion along the incline.
- Use momentum (and energy, since elastic) for the collision.
- Use work-energy or kinematics with friction for stopping distance.